

100% GEMMA

GemmaCaption

Multi-Style Video Captioning

Four voices for one video — every caption fact-checked against the frames.

formal

sarcastic

humorous_tech

humorous_non_tech

THE TASK

Four voices, one video

For every clip, GemmaCaption writes four captions of the SAME scene — each in a distinct voice.

The judge scores exactly two things:

Accuracy — faithful to what is on screen.

Style match — nails the requested tone.

formal

Professional, factual news-wire tone.

sarcastic

Dry, deadpan, lightly mocking.

humorous_tech

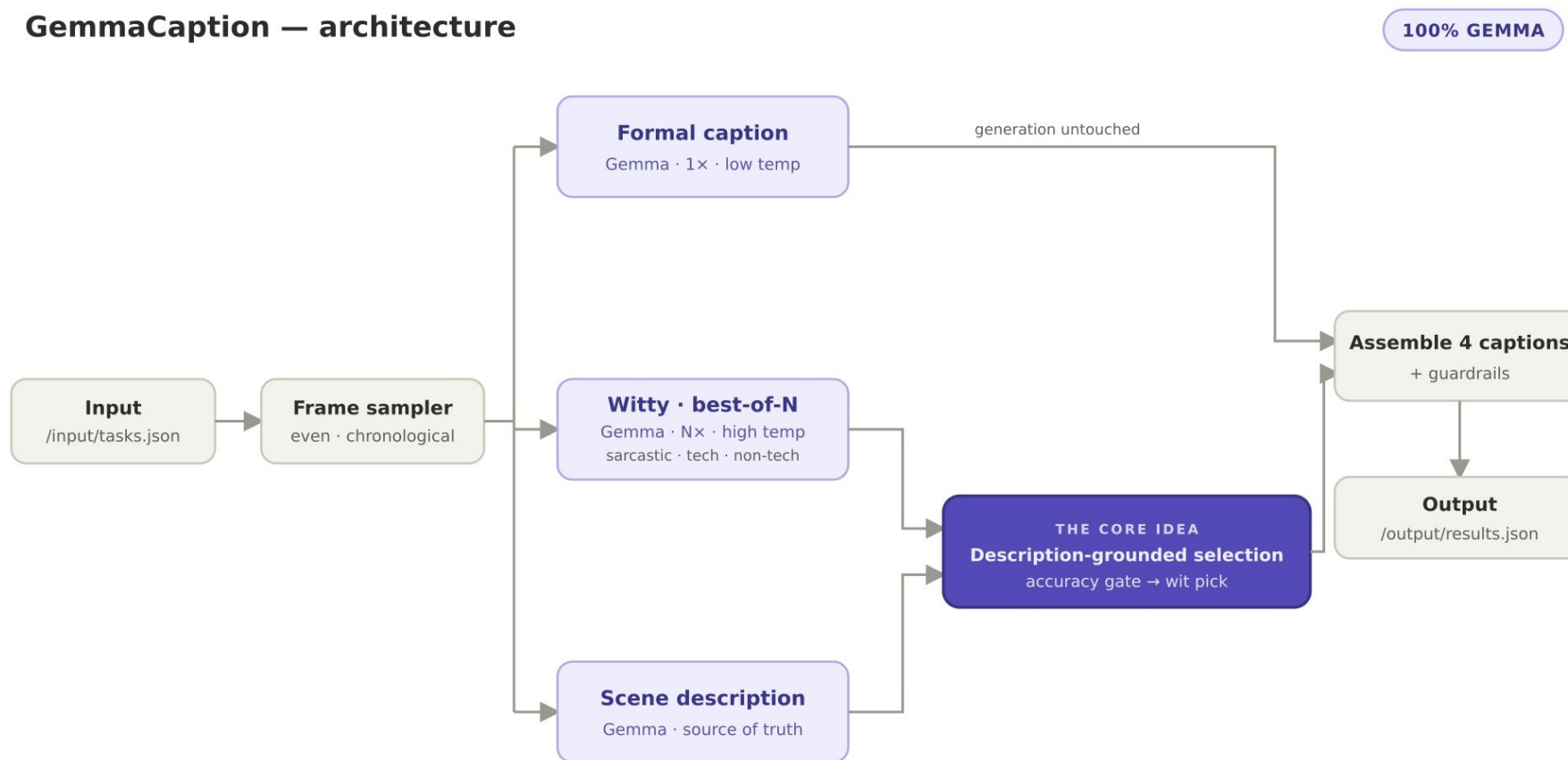
Funny, with a software / dev metaphor.

humorous_non_tech

Everyday humor, zero tech words.

An all-Gemma pipeline

GemmaCaption — architecture



Guardrails: grounding · non-tech word ban · coherence repair

Fallback: Gemma → serverless backup → templates — output is never empty

 Gemma step

 I/O · assembly

Four hand-tuned personas

formal

News-wire / stock-footage register. Present tense, third person, every clause an independently checkable fact.

sarcastic

Deadpan irony aimed at one real thing on screen. The facts stay true; only the attitude bites.

humorous_tech

One sustained software metaphor — deploys, merge conflicts, latency — mapped onto the actual action.

humorous_non_tech

Warm everyday comedy: anthropomorphize, exaggerate tiny stakes. A hard ban on any technology word.

Best-of-N + description-grounded selection

1

Draft N candidates

Gemma writes several witty captions at high temperature.

2

Ground the truth

Gemma writes a factual scene description — the source of truth.

3

Two-criterion pick

Accuracy gate then wit pick: drop hallucinations, keep the boldest.

Live example — the accuracy gate catches a hallucination the frames don't support:

REJECTED

“... a collection of cars and a **blue bus** ...” — the clip shows a red bus.

SELECTED

“... a handful of cars... vibrant yellow autumn trees...” — accurate, and still the wittiest draft.

Guardrails & reliability

Grounding rules

Never invent names, brands, counts, or on-screen text.

Non-tech word ban

Regenerates humorous_non_tech if a tech word slips in.

Coherence repair

Catches high-temp garbling and regenerates once.

Fallback chain

Gemma to serverless to templates. Output is never empty.

Complete at t=0

A valid results.json is written up front, then overwritten.

Time watchdog

Every run stays under the harness time cap.

BEST USE OF GEMMA

One model. Four roles.

Every model call in the pipeline is Gemma-4-31B — no other model touches the captions.

Perceive

reads the frames

Gemma

Write

drafts every style

Gemma

Describe

builds the source of truth

Gemma

Select

picks the best caption

Gemma

4 roles to **1 model** — and Gemma writes the final captions, so it stays prize-eligible.

Same office scene, four takes

formal

"A woman with dark hair in a high bun sits at a white desk in a modern office, typing on a white keyboard with pink-manicured nails; a potted plant sits behind her."

sarcastic

"A truly riveting display of human productivity: she engages in a high-stakes staring contest with her monitor, managing the Herculean task of tapping a few keys."

humorous_tech

"A developer mid production-incident — her face shifts from 'this should work' to 'why is it doing that' as she attempts a hotfix at the keyboard."

humorous_non_tech

"She stares at her desk with a look of pure, unadulterated betrayal, trying to make sense of whatever is happening in front of her."

SHIP IT

One container, two files

Single Docker image (linux/amd64), fully self-contained.

Reads `/input/tasks.json`, writes `/output/results.json`.

Downloads each video itself and samples frames.

Keys baked at build time — the judge injects nothing.

Always complete — a caption for every style, every task.

```
# input  /input/tasks.json
[{ "task_id": "clip-001",
  "video_url": "https://...",
  "styles": ["formal", ...] }]

# output /output/results.json
[{ "task_id": "clip-001",
  "captions": {
    "formal": "...",
    "sarcastic": "...",
    "humorous_tech": "...",
    "humorous_non_tech": "..." } }]
```

GemmaCaption

Four voices. One model. Zero hallucinated buses.

formal

sarcastic

humorous_tech

humorous_non_tech

Thank you · AMD Hackathon Track 2 · Best Use of Gemma